Lucas Wu

Electrical Engineering | Available Jan–Apr 2027 | Waterloo, ON / Calgary, AB

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EDUCATION

University of Waterloo Sep 2026–Apr 2031 (expected)
Bachelor of Applied Science candidate, Electrical Engineering (Honours, Co-op) | First year

Western Canada High School | Admission average: 96% Jun 2026

TECHNICAL SKILLS

Digital design: SystemVerilog, Verilog, cocotb, Icarus Verilog
Programming and tools: C++, Python, Java, JavaScript, HTML/CSS; Git, NumPy, OpenCV, MediaPipe, Flask
Engineering tools: Altium Designer, MATLAB

DESIGN TEAMS

UWASIC | RTL Design Member, Ray-Tracing ASIC | Portfolio Sep 2026–Present

- Wrote the functional-unit control module in SystemVerilog, decoding 11 micro-ops and dispatching them to the ALU, multiplier, and CORDIC units over valid/ready handshakes, with operand muxing and fixed-priority writeback.
- Supporting the verification team as the module is tested and integrated into the chip's RTL-to-GDS flow.
- Onboarding: built an SPI Mode 0 peripheral for Tiny Tapeout (5 registers, 16 outputs, two-stage CDC) that drives a 3 kHz PWM, and simulated it with cocotb and Icarus Verilog (GitHub).

UWAFT (UW Alternative Fuels Team) | Electrical Subteam Member | Portfolio Sep 2026–Present

- Designed a Li-ion battery monitor and relay switch board in Altium (STC3100, STM32), now being fabricated.
- Laid out the 2-layer PCB with 1.5 mm power traces, a Kelvin-sensed current shunt, and a solid ground plane.

TECHNICAL PROJECTS

SignEd | Hack the Bias, 1st Place Beginner Tier | GitHub | Portfolio Mar 2026

- Built an ASL learning website (Python, Flask, JavaScript, HTML/CSS) as the sole programmer on a 5-person team.
- Recognized all 24 static ASL letters with MediaPipe/OpenCV and distance-weighted KNN; normalized 21 3D hand landmarks by wrist position and hand size.

C++ Ray Tracer | Personal Project | Portfolio May 2025

- Built a path tracer in C++ based on *Ray Tracing in One Weekend*, implementing ray–sphere intersection, surface normals, and Monte Carlo diffuse shading with multi-sample anti-aliasing.

VEX Robotics | Team 2088D | Portfolio 2022–2024

- Designed and built the disc intake and flywheel launcher and programmed the autonomous routines; iteratively tuned flywheel speed and intake spacing to improve shooting consistency and reduce jams.
- Team placed second in Alberta Provincial Robot Skills; competed at the 2023 VEX World Championship in Dallas.

EXPERIENCE AND LEADERSHIP

Coding Instructor | Code Ninjas May–Aug 2026

- Taught game development to 15 students aged 6–15 using block-based coding and JavaScript in Microsoft MakeCode Arcade; debugged student code one-on-one and adapted lessons to each student's level.

Co-Founder and Web Designer | Calgary Ace Learning Center | Portfolio Mar 2024–Present

- Co-founded a student-run tutoring nonprofit; contributed 400+ volunteer hours tutoring and organizing, and led fundraising that raised $5,300+ for the Alberta Children's Hospital Foundation.
- Built the organization's WordPress website with booking, team, and bulletin pages.

President | Western Canada High School Computer Science Club Sep 2024–Jun 2026

- Led weekly competitive programming sessions for a 20+ member club and created the teaching materials.

ADDITIONAL EXPERIENCE AND ACHIEVEMENTS

Distinction: Canadian Computing Competition (Junior 2022, Senior 2023) | Canadian Senior Mathematics Contest (2022)
Level 2 Hockey Referee (Oct 2023–Mar 2026) | 11 seasons of competitive hockey | English and Mandarin